

Clo(o)k: Human-Time Interactions Through a Clock That “Looks”

Zhuoyue Lyu*

Department of Engineering
University of Cambridge
Cambridge, United Kingdom
zl536@cam.ac.uk

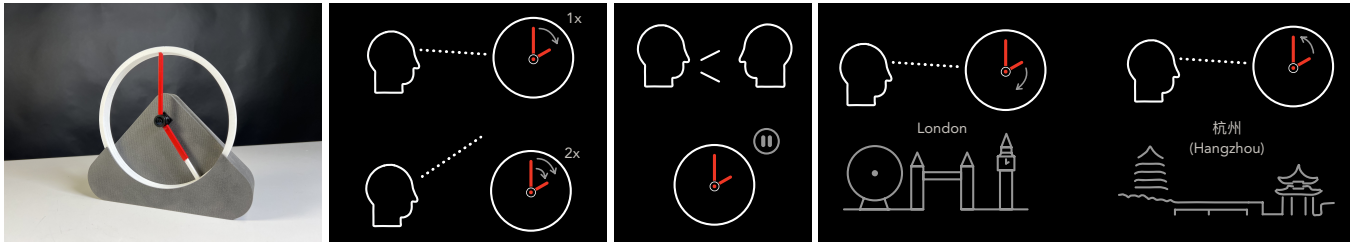


Figure 1: Clo(o)k and its three interaction types (Section 2). Video demo, design files, and code available at zhuoyuelyu.com/clook

Abstract

What if a clock could do more than tell time—what if it could look around? This project explores the conceptualization, design, and construction of a timepiece with visual perception capabilities, featuring three types of human-time interactions. Informal observations during a demonstration highlight its unique user experiences.

CCS Concepts

• Human-centered computing → Interaction devices.

Keywords

clock, time, tangible, telepresence

ACM Reference Format:

Zhuoyue Lyu. 2025. Clo(o)k: Human-Time Interactions Through a Clock That “Looks”. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3706599.3721343>

1 Introduction and Related Work

Time is a multifaceted concept, comprising both objective and subjective elements. Objectively, it is precisely measured by the ticking of clocks, while subjectively it can slip by unnoticed when we are deeply engrossed in our activities. On a global scale, time unites us: when it is 3 AM in Boston, it is simultaneously 3 PM in Beijing¹—an understanding that enables seamless communication across time zones. In popular culture, Disney films such as *Snow*

*White and the Seven Dwarfs*² and *Beauty and the Beast*³ imagine everyday objects with human personalities. Inspired by these portrayals, we wondered what would happen if a clock could see and interact with people. This notion opens creative possibilities for playful, human-clock interactions [10], motivating the design of the Clo(o)k—a clock that looks.

Previous research has explored practical uses of everyday objects [3, 9, 11]. Specifically, clocks have been employed to visualize upcoming events [2], track sleep patterns [4], and support planning and reflection [6, 7]. This work takes a more playful approach by creating an interactive experience that allows users to reflect on the concept of time. Telepresence technology has used various sensory modalities, including sound [12], touch [1, 8], and sight [5] to enable people to feel the presence of others even when they are far apart. Building on this, we integrate time into interactions, enabling new possibilities across time and space.

2 Clo(o)k Interactions

The Clo(o)k operates like a normal clock when the user is looking at it, but speeds up when the user’s attention shifts away (2nd subfigure in Figure 1). If the Clo(o)k notices that the user is engaged in conversation with someone, it stops moving to alleviate any concerns about time and allow for uninterrupted communication (3rd subfigure in Figure 1). If the user’s loved one resides in a different time zone (e.g., London, UK, and Hangzhou, China), when both parties look at their respective Clo(o)ks simultaneously, they will see each other’s current local time (4th subfigure in Figure 1).

3 Design and Fabrication

To achieve a simple, clean design for Clo(o)k, basic shapes such as circles and triangles were chosen for the clock face and base. Positioning the camera at the center preserves the appearance of a standard clock, allowing users to initially perceive it as an ordinary timepiece. The twist emerges when they realize the central

*This project was completed while the author was a student at MIT.

¹During U.S. daylight saving time (March–November).

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3721343>

²<https://movies.disney.com/snow-white-and-the-seven-dwarfs>

³<https://movies.disney.com/beauty-and-the-beast>



Figure 2: Design and fabrication of the Clo(o)k, along with its demo at the MAS.863 Open House (Dec. 20, 2022, MIT Media Lab).

element is a camera. However, this placement precluded the use of a traditional clock mechanism, where all hands attach at the center. Instead, the hollow clock mechanism⁴ was adopted, with the minute and hour hands integrated into rings driven by concealed gear sets in the base (Figure 2). The 3D design was created using Fusion 360⁵ and Blender⁶, and then exported as STL files to be printed. The rings and gears were printed with PLA using Prusa⁷, while the hands were spray-painted red. The base was printed using Fuse 1⁸ with powder for better quality.

4 Hardware and Software

Clo(o)k utilizes an ESP32-CAM⁹ with a modified lens to capture the video, control one of the stepper motors, and wirelessly communicate with the computer and other Clo(o)ks. A D11C¹⁰ board was designed and milled to extend the pins of the ESP32-CAM, enabling it to receive serial signals from the ESP32-CAM and control another stepper motor. Two 28BYJ-48 stepper motors¹¹ with driver boards control the rings. An FTDI Converter¹² programs the ESP32-CAM and supplies power (Figure 2). Both ESP32-CAM and D11C were programmed using Arduino¹³. Although there is a reduced version of OpenCV¹⁴ for ESP32-CAM, for more flexibility and convenience, a separate laptop running Python script using OpenCV¹⁵ is used for facial detection, which receives the video stream from ESP32-CAM wirelessly. ESP32-CAM updates the movements of the stepper motors based on the information (number of faces) it receives from Python through the USB serial connection.

5 Informal Observations and Limitations

Clo(o)k was demonstrated during an Open House at the MIT Media Lab (Figure 2). Attendees consistently responded positively upon learning about the Clo(o)k's interactions, particularly appreciating the exploration of time's dual nature (subjective and objective). Many attendees were drawn to the minimalist design and expressed pleasant surprise upon discovering the detachable ring mechanism. Several international students with loved ones in different time zones highlighted the usefulness of the third interaction, noting its ability to evoke a sense of connection with distant family and friends. However, some limitations were identified: low lighting hindered facial recognition accuracy, and latency in serial communication between the ESP32-CAM and D11C caused slight delays in the minute and hour hands' movements.

6 Conclusion and Future Work

We designed Clo(o)k, a clock system that enables playful human-time interactions, suggesting a novel use of tangible, everyday objects to engage with intangible concepts. The demonstrated interactions represent only a subset of its potential uses. The detachable rings and programmable motors enable customization (Figure 3), inviting further exploration within and beyond the HCI community.



Figure 3: Potential applications of the Clo(o)k system. The programmable rings can carry elements beyond hour and minute hands: (1) Planetary Movement – independently rotating rings carry planets (Earth and Saturn) around a central Sun. (2) Interactive Face – a fixed front ring with eyes remains static, while the back ring with various attached mouth shapes rotates to show different facial expressions based on camera input. (3) Kinetic Art – two rings with dynamic patterns rotate at varying speeds and directions, creating mesmerizing visuals.

⁴<https://www.thingiverse.com/thing:4761858>

⁵<https://www.autodesk.com/products/fusion-360/>

⁶<https://www.blender.org/>

⁷<https://www.prusa3d.com/>

⁸<https://formlabs.com/3d-printers/fuse-1/>

⁹<https://docs.ai-thinker.com/en/esp32-cam>

¹⁰https://ww1.microchip.com/downloads/en/DeviceDoc/Atmel-42363-SAM-D11_Summary.pdf

¹¹<https://components101.com/motors/28byj-48-stepper-motor>

¹²<https://microcontrollerslab.com/ftdi-usb-to-serial-converter-cable-use-linux-windows/>

¹³<https://www.arduino.cc/>

¹⁴<https://github.com/joachimBurket/esp32-opencv>

¹⁵<https://opencv.org/>

Acknowledgments

We thank Dr. Marcelo Coelho, Bill McKenna, Dominic Lim Co, and the students of MIT 4.031 for their valuable feedback on the design and fabrication of the Clo(o)k. We are grateful to Prof. Neil Gershenfeld, the TAs, and students of MAS.863 for their guidance and support on the electronics. We appreciate Prof. Hiroshi Ishii and his Tangible Interfaces class for inspiring the interaction design. Finally, we extend our thanks to everyone who visited our booth at the Open House of MAS.863 for their insightful comments and suggestions.

References

- [1] Scott Brave and Andrew Dahley. 1997. InTouch: A Medium for Haptic Interpersonal Communication. In *CHI '97 Extended Abstracts on Human Factors in Computing Systems* (Atlanta, Georgia) (CHI EA '97). Association for Computing Machinery, New York, NY, USA, 363–364. <https://doi.org/10.1145/1120212.1120435>
- [2] Pierre Dragicevic and Stéphane Huot. 2002. SpiraClock: A Continuous and Non-Intrusive Display for Upcoming Events. In *CHI '02 Extended Abstracts on Human Factors in Computing Systems* (Minneapolis, Minnesota, USA) (CHI EA '02). Association for Computing Machinery, New York, NY, USA, 604–605. <https://doi.org/10.1145/506443.506505>
- [3] Cathy Mengying Fang, Ryo Suzuki, and Daniel Leithinger. 2023. VR Haptics at Home: Repurposing Everyday Objects and Environment for Casual and On-Demand VR Haptic Experiences. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 312, 7 pages. <https://doi.org/10.1145/3544549.3585871>
- [4] Fabian Hemmert, Susann Hamann, and Reto Wettach. 2009. The Digital Hourglass. In *Proceedings of the 3rd International Conference on Tangible and Embedded Interaction* (Cambridge, United Kingdom) (TEI '09). Association for Computing Machinery, New York, NY, USA, 19–20. <https://doi.org/10.1145/1517664.1517672>
- [5] Hiroshi Ishii and Minoru Kobayashi. 1992. ClearBoard: A Seamless Medium for Shared Drawing and Conversation with Eye Contact. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Monterey, California, USA) (CHI '92). Association for Computing Machinery, New York, NY, USA, 525–532. <https://doi.org/10.1145/142750.142977>
- [6] Sarah Janböcke, Alina Gawlitta, Judith Dörrenbächer, and Marc Hassenzahl. 2020. Finding the Inner Clock: A Chronobiology-Based Calendar. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–7. <https://doi.org/10.1145/3334480.3382830>
- [7] Kyung-Ryong Lee, Somi Ju, Temirlan Dzhoroiev, Geonil Goh, Moon-Hwan Lee, and Young-Woo Park. 2020. DayClo: An Everyday Table Clock Providing Interaction with Personal Schedule Data for Self-Reflection. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference* (Eindhoven, Netherlands) (DIS '20). Association for Computing Machinery, New York, NY, USA, 1793–1806. <https://doi.org/10.1145/3357236.3395439>
- [8] Daniel Leithinger, Sean Follmer, Alex Olwal, and Hiroshi Ishii. 2014. Physical Telepresence: Shape Capture and Display for Embodied, Computer-Mediated Remote Collaboration. In *Proceedings of the 27th Annual ACM Symposium on User Interface Software and Technology* (Honolulu, Hawaii, USA) (UIST '14). Association for Computing Machinery, New York, NY, USA, 461–470. <https://doi.org/10.1145/2642918.2647377>
- [9] Zhuoyue Lyu and Per Ola Kristensson. 2025. Objestures: Bimanual Interactions with Everyday Objects and Mid-Air Gestures in Mixed Reality. arXiv:2503.02973 [cs.HC] <https://arxiv.org/abs/2503.02973>
- [10] Joseph E McGrath and Janice R Kelly. 1986. *Time and human interaction: Toward a social psychology of time*. Guilford Press.
- [11] Kyzyl Monteiro, Ritik Vatsal, Neil Chulpongsatorn, Aman Parnami, and Ryo Suzuki. 2023. Teachable Reality: Prototyping Tangible Augmented Reality with Everyday Objects by Leveraging Interactive Machine Teaching. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 459, 15 pages. <https://doi.org/10.1145/3544548.3581449>
- [12] Xiao Xiao and Hiroshi Ishii. 2010. MirrorFugue: Communicating Hand Gesture in Remote Piano Collaboration. In *Proceedings of the Fifth International Conference on Tangible, Embedded, and Embodied Interaction* (Funchal, Portugal) (TEI '11). Association for Computing Machinery, New York, NY, USA, 13–20. <https://doi.org/10.1145/1935701.1935705>